1) **Write a ‘C’ program to find minimum of two numbers.**
Ans -

```
#include <stdio.h>
int main()
{
   int a, b;

   printf("Enter two numbers: ");
   scanf("%d %d", &a, &b);

   if (a < b)
      printf("Minimum = %d", a);
   else
      printf("Minimum = %d", b);

   return 0;
}
```

Input- Enter two numbers: 10 5
Output - Minimum = 5

2) **Write a ‘C’ program to accept an array of n float values and display them in the reverse order**
Ans -

```
#include <stdio.h>
int main()
{
   int n, i;
   float a[100];

   printf("Enter number of elements: ");
   scanf("%d", &n);

   printf("Enter float values:\n");
   for (i = 0; i < n; i++)
   {
      scanf("%f", &a[i]);
   }

   printf("Array in reverse order:\n");
   for (i = n - 1; i >= 0; i--)
   {
      printf("%.2f ", a[i]);
   }
```

```
    return 0;
}
```

I/P-Enter number of elements: 5
Enter float values:
1.5 2.5 3.5 4.5 5.5
o/p - Array in reverse order:
5.50 4.50 3.50 2.50 1.50

3) **Write a 'C' program with a menu to perform the following operations on a character. 1. Check uppercase or lowercase 2. Display its ASCII value 3. Display its next and previous character 4. Exit .**

Ans -

```
#include <stdio.h>
  int main()
{
    char ch;
    int choice;

    printf("Enter a character: ");
    scanf(" %c", &ch);

    do
    {
        printf("\n1. Check uppercase or lowercase");
        printf("\n2. Display ASCII value");
        printf("\n3. Display next and previous character");
        printf("\n4. Exit");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                if (ch >= 'A' && ch <= 'Z')
                    printf("Uppercase");
                else if (ch >= 'a' && ch <= 'z')
                    printf("Lowercase");
                else
                    printf("Not an alphabet");
                break;

            case 2:
                printf("ASCII value = %d", ch);
                break;
```

```
            case 3:
                printf("Next character = %c", ch + 1);
                printf("\nPrevious character = %c", ch - 1);
                break;

            case 4:
                printf("Exit");
                break;

            default:
                printf("Invalid choice");
        }
    } while (choice != 4);

    return 0;
}
```

O/P-Enter a character: A

1. Check uppercase or lowercase
2. Display ASCII value
3. Display next and previous character
4. Exit
Enter your choice: 1
Uppercase

Enter your choice: 2
ASCII value = 65

Enter your choice: 3
Next character = B
Previous character = @

Enter your choice: 4
Exit

4) **Write a 'C' program to check whether the given year is leap year or not.**

Ans -

```
#include <stdio.h>
int main()
{
    int year;

    printf("Enter a year: ");
```

```
    scanf("%d", &year);

    if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
        printf("Leap year");
    else
        printf("Not a leap year");

    return 0;
}
```

I/P -Enter a year: 2024
O/P- Leap year

5) Write a menu driven program to perform the following operations on an integer. Write separate functions. 1. Check number is even or odd 2. Check number is prime or not 3. Exit

Ans -

```
#include <stdio.h>
void evenOdd(int n)
{
    if (n % 2 == 0)
        printf("Even\n");
    else
        printf("Odd\n");
}

void prime(int n)
{
    int i;

    if (n < 2)
    {
        printf("Not Prime\n");
        return;
    }

    for (i = 2; i < n; i++)
    {
        if (n % i == 0)
        {
            printf("Not Prime\n");
            return;
        }
    }
```

```
    printf("Prime\n");
}

int main()
{
    int n, ch;

    printf("Enter number: ");
    scanf("%d", &n);

    do
    {
        printf("\n1.Even/Odd\n2.Prime\n3.Exit\n");
        scanf("%d", &ch);

        switch (ch)
        {
            case 1: evenOdd(n); break;
            case 2: prime(n); break;
            case 3: break;
            default: printf("Invalid choice\n");
        }
    } while (ch != 3);

    return 0;
}

)
```

I/P- Enter number: 7
1.Even/Odd
2.Prime
3.Exit
1
O/P -Odd

6) Write a recursive function in ‘C’ to calculate the factorial of a number.

Ans-

```
#include <stdio.h>
int fact(int n)
{
    if (n == 0)
        return 1;
    return n * fact(n - 1);
}
```

```
int main()
{
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    printf("Factorial = %d", fact(n));

    return 0;
}
```

I/P - Enter a number: 5
O/P - Factorial = 120